SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS - Final answers with no solution will get ZERO

Carry your intermediate computations to at least three decimal places.

Q1- *Use the golden-section search to find the minimum of* $y = x^2 - 4\sin(5x)$ *assume that* $x_l = 1.3$ *and* $x_u = 1.7$ *(10 marks)*

Answer:

.....

Q2- Solve the integral $\int_0^1 \frac{3}{1+x^2} dx$ by Trapezoidal rule with $n = 1, 2, 4$ and 8. Then obtain the better estimate using **Romberg integration** method. *(13 marks)*

Answer:

.....

Q3- Consider the ordinary differential equation: $2\frac{dy}{dx} - 3e^x + 2y = 0$

Given: $y(0) = 0, h = 0.5$. Use the Runge-Kutta method to the 4th order to find $y(1)$. *(12 marks)*

Answer:

.....

Q4:From the following data shown in the table below (15 marks)

x:	-1	0	1	3	5
f(x)	3.1	0	4.3	7.5	2.2

1- Find $f(2)$- employing Newton's forward interpolation formula.

Answer:

.....

2-Find the area bounded by the curve and x axis from x= -1 to x=5 using Simpson's 1/3 rule

Answer:

.....

3- Find $f'(3)$ -using two points Centered finite-divided difference formula taking the data from the table.

Answer:.....

End of questions

⇒ (Q.2)

ملاحظات :- بسؤال مش محدد لازم توقف و بس بالإمتحان محاضر ☺ (تيريشن)
إعملوا و كان ناسي ☺ = سيبه

⇒ sol :-

$y \equiv f(x) = x^2 - 4\sin(5x)$ / $[X_L = 1.3 \ , \ X_u = 1.7]$

(minimum) المطلوب

(Rad) ← الآلة راد
بلاش تشوفوا على الحديقة

⇒ $d = 0.61803\,(X_u - X_L) \Rightarrow d = 0.61803\,(1.7 - 1.3)$

⇒ $\boxed{d = 0.247}$ ✱

⇒ $X_1 = X_L + d \Rightarrow X_1 = 1.3 + 0.247 \Rightarrow \boxed{X_1 = 1.547}$ ✱

⇒ $X_2 = X_u - d \Rightarrow X_u = 1.7 - 0.247 \Rightarrow \boxed{X_2 = 1.453}$ ✱

⇒ $f(X_1) = (1.547)^2 - [4\sin(5*1.547)] \Rightarrow \boxed{f(X_1) = -1.589}$ ✱

⇒ $f(X_2) = (1.453)^2 - [4\sin(5*1.453)] \Rightarrow \boxed{f(X_2) = -1.215}$ ✱

لأن المطلوب minimum ⇒ $f(X_2) > f(X_1) \Rightarrow \underline{X_L = X_2}$ & $\underline{X_u = X_u}$] ✱ من النتيجة

⇒ new interval → $\boxed{X_L = 1.453 \ , \ X_u = 1.7}$ ✱ → هيا لخلصنا 2 iteration

✱ ✱ ✱ ✱ ✱ ✱

⇒ $d = 0.61803\,(X_u - X_2) \Rightarrow d = 0.61803\,(1.7 - 1.453)$

⇒ $\boxed{d = 0.153}$ ✱

⇒ $X_1 = X_L + d \Rightarrow X_1 = 1.453 + 0.153 \Rightarrow \boxed{X_1 = 1.606}$ ✱

⇒ $X_2 = X_u - d \Rightarrow X_2 = 1.7 - 0.153 \Rightarrow \boxed{X_2 = 1.547}$ ✱

⇒ $f(X_1) = (1.606)^2 - [4\sin(5*1.606)] \Rightarrow \boxed{f(X_1) = -1.359}$ ✱

⇒ $f(X_2) = (1.547)^2 - [4\sin(5*1.547)] \Rightarrow \boxed{f(X_2) = -1.579}$ ✱

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min ⇒ ⇒ [f(X1) > f(X2) ⇒ Xu = X1 & Xl = Xl] ✳

من الشرط

⇒ New interval → [Xu = 1.606 , Xl = 1.453] ✳

⇒ d = 0.61803 (Xu - Xl) ⇒ d = 0.61803 (1.606 - 1.453)

⇒ d = 0.095 ✳

⇒ X1 = Xl + d ⇒ X1 = 1.453 + 0.095 ⇒ X1 = 1.548 ✳

⇒ X2 = Xu − d ⇒ X2 = 1.606 - 0.095 ⇒ X2 = 1.511 ✳

⇒ f(X1) = (1.548)² - [4 sin(5 * 1.548)] ⇒ f(X1) = -1.578 ✳

⇒ f(X2) = (1.511)² - [4 sin(5 * 1.511)] ⇒ f(X2) = -1.539 ✳

min ⇒ ⇒ [f(X2) > f(X1) ⇒ Xl = X2 & Xu = Xu] ✳

من الشرط

⇒ New interval ⇒ [Xl = 1.511 , Xu = 1.606]

Ans = ✳

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[Q.2]. $\left[\int_0^1 \frac{3}{1+x^2}\cdot dx\right]$

تساوي ← عدد الأرقام - 1 ✳

⇒ sol:-

$f(x) = \frac{3}{1+x^2}$ / $n = 1, 2, 4, 8 \Rightarrow 3^{rd}$ order.

(i,k)	2nd	2nd	3rd
R(1,0)	[illegible]	—	—
R(2,0)	R(2,1)	[illegible]	—
R(3,0)	R(3,1)	R(3,2)	—
R(4,0)	R(4,1)	R(4,2)	R(4,3)

جاي لين لازم نوصلها

:)

⇒ n = 1.

$\Rightarrow h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{1} = \boxed{1}$ ✳

$\Rightarrow R(1,0) = \frac{h}{2}[f(0) + f(1)]$

$= \frac{1}{2}[3 + 1.5]$

$\Rightarrow \boxed{R(1,0) = 2.25}$ ✳

⇒ n = 2.

$\Rightarrow h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{2} = \boxed{0.5}$ ✳

$\Rightarrow R(2,0) = \frac{h}{2}[f(0) + f(1) + (2 * f(0.5)]$

$= \frac{0.5}{2}[3 + 1.5 + (2 * 2.4)]$

$\boxed{R(2,0) = 2.325}$ ✳

⇒ n = 4.

$\Rightarrow h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{4} = \boxed{0.25}$ ✳

$\Rightarrow R(3,0) = \frac{h}{2}[f(0) + f(1) + [2(f(0.25) + f(0.5)] + f(0.75)]$

$\Rightarrow R(3,0) = \frac{0.25}{2}[3 + 1.5 + [2(2.823 + 2.4)] + 1.92]$

$\Rightarrow \boxed{R(3,0) = 2.348}$ ✳

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$\Rightarrow n = 8$

$\Rightarrow h = \frac{b-a}{n} \Rightarrow h = \frac{1-0}{8} = \boxed{0.125}$ ✱

$\Rightarrow R(4,0) = \frac{h}{2}\left[f(0)+f(3)+2\left(f(0.125)+f(0.25)+f(0.375)\right.\right.$

$\left.\left.+ f(0.5)+f(0.625)+f(0.75)+f(0.875)\right)\right]$

$\Rightarrow R(4,0) = \frac{0.125}{2}\left[3+1.5+2(2.954+2.823+2.63+\right.$

$\left.+ 2.4+2.157+1.92+1.699)\right]$

$\Rightarrow \boxed{R(4,0) = 2.354}$ ✱

⇐ هنا نطبق كل قانون Rom ← من الشيت.

$\Rightarrow R^{i}_{k}(2,1) = \frac{(4^1 * R(2,0)) - R(1,0)}{4^1 - 1} \Rightarrow R(2,1) = \frac{(4*2.325)-(2.75)}{3}$

$\Rightarrow \boxed{R(2,1) = 2.35}$ ✱

$\Rightarrow R(3,1) = \frac{4^1 * R(3,0) - R(2,0)}{4^1 - 1} \Rightarrow R(3,1) = \frac{(4*2.348)-(2.325)}{3}$

$\Rightarrow \boxed{R(3,1) = 2.356}$ ✱

$R(4,1) = \frac{4^1 * R(4,0) - R(3,0)}{4^1 - 1} \Rightarrow R(4,0) = \frac{(4*2.354)-2.348}{3}$

$\Rightarrow \boxed{R(4,1) = 2.356}$ ✱

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$$\Rightarrow R(3,2) = \frac{(4^2 * R(3,1)) - R(2,1)}{4^2 - 1}$$

$$\Rightarrow R(3,2) = \frac{(16 * 2.356) - 2.35}{15} \Rightarrow \boxed{R(3,2) = 2.356}$$

$$\Rightarrow R(4,2) = \frac{(4^2 * R(4,1)) - R(3,1)}{15}$$

$$\Rightarrow R(4,2) = \frac{(16 * 2.356) - (2.356)}{15} \Rightarrow \boxed{R(4,2) = 2.356}$$

$$\Rightarrow R(4,3) = \frac{(4^3 * R(4,2)) - R(3,2)}{63}$$

$$\Rightarrow R(4,3) = \frac{(64 * 2.356) - 2.356}{63}$$

$$\Rightarrow R(4,3) = \boxed{2.356}$$

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[Q.3].

⇒ sol :-

$f(x_i, y_i)$

$$\frac{2dy}{dx} - 3e^x + 2y = 0 \Rightarrow \left[\frac{dy}{dx} = \frac{3e^x - 2y}{2}\right]$$

$y(0) = 0$ ← $x_i = 0$, $y_i = 0$ / $h = 0.5$ / $y(1)$

$y(0.5)$
$y(1)$] (2 step) لحد ← المطلوب ← [RK4]

$$\Rightarrow k_1 = h f(x_0, y_0) \Rightarrow k_1 = 0.5 f(0,0) \Rightarrow k = 0.5 * 1.5 \Rightarrow \boxed{k_1 = 0.75} *$$

$$\Rightarrow k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow k_2 = 0.5 f(0.25, 0.375) \Rightarrow k_2 = 0.5 * 1.551$$

$$\Rightarrow \boxed{k_2 = 0.775} *$$

$$\Rightarrow k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \Rightarrow k_3 = 0.5 f(0.25, 0.387) \Rightarrow k_3 = 0.5 * 1.539$$

$$\Rightarrow \boxed{k_3 = 0.769} *$$

$$\Rightarrow k_4 = h f(x_0 + h, y_0 + k_3) \Rightarrow k_4 = 0.5 f(0.5, 0.769) \Rightarrow k_4 = 0.5 * 1.704$$

$$\Rightarrow \boxed{k_4 = 0.852} *$$

(y_{i+1}) (y_i)

$$\Rightarrow y(0.5) = y(0) + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$\Rightarrow y(0.5) = 0 + \frac{1}{6}(0.75 + (2 * 0.775) + (2 * 0.769) + 0.852)$$

$$\boxed{y(0.5) = 0.782} * \Rightarrow \text{step (1)} \rightarrow \underline{\text{done}}$$

*

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$\Rightarrow$ step (2) $\rightarrow [x_0 = 0.5 \;/\; y_0 = 0.782]$ ✳

$\Rightarrow k_1 = h f(x_0, y_0) \Rightarrow k_1 = 0.5 f(0.5, 0.782) \Rightarrow k_1 = 0.5 * 1.691$

$\Rightarrow \boxed{k_1 = 0.845}$ ✳

$\Rightarrow k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow k_2 = 0.5 f(0.75, 1.204)$

$\Rightarrow k_2 = 0.5 * 1.971 \Rightarrow \boxed{k_2 = 0.985}$ ✳

$\Rightarrow k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \Rightarrow k_3 = 0.5 f(0.75, 1.274)$

$\Rightarrow k_3 = 0.5 * 1.901 \Rightarrow \boxed{k_3 = 0.950}$ ✳

$\Rightarrow k_4 = h f(x_0 + h, y_0 + k_3) \Rightarrow k_4 = 0.5 f(1, 1.732)$

$\Rightarrow k_4 = 0.5 * 2.345 \Rightarrow \boxed{k_4 = 1.172}$ ✳

$\Rightarrow y(1) = y(0.5) + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$\Rightarrow y(1) = 0.782 + \frac{1}{6}(0.845 + (2 * 0.985) + (2 * 0.950) + 1.172)$

$\Rightarrow \boxed{y(1) = 1.730}$ ✳ $\Rightarrow$ step (2) $\rightarrow$ done

✳

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(Q.4)

x	-1	0	1	3	5
f(x)	3.1	0	4.3	7.5	2.2

(1). ⇒ sol:-

x_i	y_j	b_1	b_2	b_3	b_4
-1	3.1	-3.1	3.7	-1.15	0.186
0	0	4.3	-0.9	-0.032	
1	4.3	1.6	-1.062		
3	7.5	-2.65			
5	2.2				

ملاحظات للتذكير:-
- عندي (5) نقاط إذا وقف عند الدرجة (4) ⇐ وبالتالي لو وقف عند ↪ (b_4)

$$\Rightarrow b_{1(1)} = \frac{3.1 - 0}{-1 - 0} = -3.1$$

$$\Rightarrow b_{1(2)} = \frac{0 - 4.3}{(5) - 1} = 4.3$$

$$\Rightarrow b_{1(3)} = \frac{4.3 - 7.5}{(1) - (3)} = 1.6$$

$$\Rightarrow b_{1(4)} = \frac{7.5 - 2.2}{3 - 5} = -2.65$$

$$\Rightarrow b_{2(1)} = \frac{-3.1 - 4.3}{-1 - 1} = 3.7$$

$$\Rightarrow b_{2(2)} = \frac{4.3 - 1.6}{0 - 3} = -0.9$$

$$\Rightarrow b_{3(3)} = \frac{1.6 \overset{(-)}{+} 2.65}{1 - 5} = -1.062$$

$$\Rightarrow b_{3(1)} = \frac{3.7 \overset{(-)}{+} 0.9}{-1 - 3} = -1.15$$

$$\Rightarrow b_{3(2)} = \frac{-0.9 \overset{(-)}{+} 1.062}{0 - 5} = -0.032$$

$$b_4 = \frac{-1.15 \overset{(-)}{+} 0.032}{-1 - 5}$$

$$b_4 = 0.186$$

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$$\Rightarrow f(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$
$$+ b_4(x - x_0)(x - x_1)(x - x_2)(x - x_3)$$

دايما زقل بواحد

$$\Rightarrow f(2) = 3.1 + [-3.1(2 + 1)] + [3.7(2 + 1)(2 - 0)]$$
$$+ [-1.15(2+1)(2-0)(2-1)] + [0.186(2+1)(2-0)(2-1)(2-3)]$$

$$\Rightarrow \boxed{f(2) = 7.984}$$

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⇒ (2).

⇒ sol:-

$X = -1 \xrightarrow{to} X = 5$

عندي (5) نقاط ← إذا ← $h = 5-1 = 4$ ← عدد زوجي

بقدر أعمل قاعدة ← sim($\frac{1}{3}$) ← بس ← h ← مش متساويات إذا لازم أكسر النقاط

⇒ $X = -1 \xrightarrow{to} X = 1$ ← عندي 3 نقاط

[-1 & 0 & 1]

ولاحظ إن (h) دون متساويات وتساوي = 1

$\Rightarrow X_0 = -1$, $X_1 = 0$, $X_2 = 1$

$\Rightarrow I_{1}_{(-1\to1)} = \frac{h}{3}[f(X_0) + f(X_2) + 4f(X_1)]$

$\Rightarrow I = \frac{1}{3}[3.1 + 4.3 + (4*0)]$

$\Rightarrow \boxed{I_1 = 2.467}$ ※

القيم بيتم أخذها من الجدول ومن ذول السؤال.

⇐ هنا ← بدي نقطتين متوخذين ← (Trap ----)

$\Rightarrow X = 3$ & $X = 5$ $\Rightarrow \boxed{h = 2}$ / $X_0 = 3$, $X_1 = 5$

$\Rightarrow I_2 = \frac{h}{2}[f(X_0) + f(X_3)]$

$= \frac{2}{2}[7.5 + 2.2]$

$= 9.7$

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$\Rightarrow I_{total} = I_1 + I_2 = 2.467 + 9.7$

$\boxed{I_{tot} = 12.167}$ ※

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(3). ⇒ solve

$f'(3)$ → use → center & first dir --- (مستقل وكمل)

$$\Rightarrow f'(3) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$$

كيف بدي أعرفها ← مدمجنا هو الجدول لأنه هو أساس السؤال ومنه الـ (h) وقدمه بنحلوا.

⇐ هو بده ← $f'(3)$ ← طيب ← الـ (3) بيني بين وبين واقعاد

	3	5
4.3	7.5	2.2

$\Rightarrow \boxed{h = 2}$

$$\Rightarrow f'(3) = \frac{f(5) - f(1)}{2 * 2} = \frac{2.2 - 4.3}{4}$$

$$\Rightarrow \boxed{f'(3) = -0.525}$$

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دعواتكم لأهلنا في غزة العزة

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